

Zhen Chen

Massachusetts Institute of Technology
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POSITIONS

Postdoctoral Associate – The Picower Institute for Learning and Memory, MIT 2023 – Present
(Advised by Earl K. Miller)

EDUCATION

Ph.D. in Brain and Cognitive Sciences – University of Rochester 2018 – 2023
(Co-advised by Krishnan Padmanabhan and Ralf Haefner)

Ph.D. in General and Fundamental Mechanics – Nanjing University of Aeronautics and
Astronautics (Advised by Xianbin Liu) 2013 – 2018

BE in Engineering Mechanics – Nanjing University of Aeronautics and Astronautics 2007 – 2011

FELLOWSHIPS & AWARDS

AS&E Graduate Student Association Conference Travel Award, University of Rochester 2021

NSF NRT: Graduate Training in Data-Enabled Research into Human Behavior and its Cognitive
and Neural Mechanisms, National Science Foundation 2018

China National Scholarship for Doctoral Students, Ministry of Education of the People's
Republic of China 2016

China National Scholarship for Undergraduate Students, Ministry of Education of the People's
Republic of China 2009

PUBLICATIONS

1. **Zhen Chen**, Scott L Brincat, Mikael Lundqvist, Roman F Loonis, Melissa R Warden, Earl K Miller. 2026. Oscillatory Control of Cortical Space as a Computational Dimension. *Current Biology* **36**: 12.
2. **Zhen Chen**, Krishnan Padmanabhan. 2025. Adult-neurogenesis allows for representational stability and flexibility in early olfactory system. *Elife* **14**.
3. **Zhen Chen**, Krishnan Padmanabhan. 2022. Top-down feedback enables flexible coding strategies in the olfactory cortex. *Cell Reports* **38**: 12.

4. **Zhen Chen**, Krishnan Padmanabhan. 2020. Top-Down Control of Inhibitory Granule Cells in the Main Olfactory Bulb Reshapes Neural Dynamics Giving Rise to a Diversity of Computations. *Frontiers in Computational Neuroscience* **14**: 59.
5. **Zhen Chen**, Jinjie Zhu, Xianbin Liu. 2019 Non-differentiability of Quasi-potential and Non-smooth Dynamics of Optimal Paths in the Stochastic Morris-Lecar Model: Type I and II Excitability. *Nonlinear Dynamics* **96** (4): 2293.
6. **Zhen Chen**, Jinjie Zhu, Xianbin Liu. 2017 Crossing the quasi-threshold manifold of a noise-driven excitable system. *Proceedings of The Royal Society A* **473**: 20170058.
7. **Zhen Chen**, Xianbin Liu. 2017 Subtle escaping modes and subset of patterns from a non-hyperbolic chaotic attractor. *Physical Review E* **95**: 012208.
8. **Zhen Chen**, Xianbin Liu. 2017 Noise induced transitions and topological study of a periodically driven system. *Communications in Nonlinear Science and Numerical Simulation* **48**: 454–461.
9. **Zhen Chen**, Xianbin Liu. 2017 Singularities of fluctuational paths for an over-damped two-well system driven by white noise. *Physica A* **469**: 206-215.
10. **Zhen Chen**, Yang Li, Xianbin Liu. 2016 Noise induced escape from a non-hyperbolic chaotic attractor of a periodically driven nonlinear oscillator. *Chaos* **26**: 935–992.
11. **Zhen Chen**, Xianbin Liu. 2016 Patterns and singular features of extreme fluctuational paths of a periodically driven system. *Physics Letters A* **380**: 1953-1958.
12. **Zhen Chen**, Xianbin Liu. 2015 Effect of bounded noise on chaotic motions of stochastically perturbed slowly varying oscillator. *Chaos, Solitons & Fractals* **76**: 72-81.
13. Jinjie Zhu, **Zhen Chen**, Xianbin Liu. 2018 Probability evolution method for exit location distribution. *Physics Letters A* **382** (11), 771-775
14. Jinjie Zhu, **Zhen Chen**, Xianbin Liu 2016 Effects of distance-dependent delay on small-world neuronal networks. *Physical Review E* **93**: 042417.
15. Chen Kong, **Zhen Chen**, Xianbin Liu. 2016 On the Stochastic Dynamical Behaviors of a Nonlinear Oscillator Under Combined Real Noise and Harmonic Excitations. *Journal of Computational and Nonlinear Dynamics* **12**: 031015.

CONFERENCES

1. Poster – **Zhen Chen**, Scott Brincat, Earl K. Miller. Oscillatory Control of Cortical Space as a Computational Dimension. *Society for Neuroscience (SfN) 2025*, November 2025, San Diego, CA USA.
2. Poster – **Zhen Chen**, Scott Brincat, Earl K. Miller. Context-dependent flow of beta and gamma waves in models of working memory tasks. *Society for Neuroscience (SfN) 2024*, October 2024, Chicago, IL USA.
3. Talk – **Zhen Chen**, Krishnan Padmanabhan. Adult-neurogenesis allows for representational stability and flexibility in early olfactory system. *Society for Neuroscience (SfN) 2023*, November 2023, Washington, D.C. USA.

4. Poster – Ying Lin, Jose Reynoso, **Zhen Chen**, Ralf Haefner, & Dujie Tadin. A new approach for the study of visual orientation perception and decisions. *Vision Science Society (VSS)* 2023, Tampa, FL USA.
5. Poster – **Zhen Chen**, Krishnan Padmanabhan. Cortical feedback regulation of adult-neurogenesis in olfactory bulb could contribute to representational flexibility of odors in olfactory piriform cortex. *Society for Neuroscience (SfN)* 2022, November 2022, San Diego, CA USA.
6. Poster – Ying Lin, **Zhen Chen**, Ralf Haefner, Dujie Tadin. Moving toward a unifying framework for perceptual decision making that combines threshold and reaction time approaches. *Vision Science Society (VSS)* 2022, May 2022, Tampa, FL USA.
7. Poster – **Zhen Chen**, Krishnan Padmanabhan. Top-down feedback gates flexible coding strategies in olfactory cortex. *Association for Chemoreception Sciences (AChemS)* 2022, April 2022, virtual.
8. Poster – **Zhen Chen**, Forrest Wargo, Rodrigo Echeveste, Ralf Haefner. Optimizing for fast sampling-based inference yields oscillatory dynamics in a spiking model of primary visual cortex. *Spiking Neural networks as Universal Function Approximators (SNUFA)* 2021, November 2021, virtual.
9. Poster – **Zhen Chen**, Forrest Wargo, Rodrigo Echeveste, Ralf Haefner. Optimizing for fast sampling-based inference yields oscillatory dynamics in a spiking model of primary visual cortex. *Bernstein Conference 2021*, September 2021, virtual.
10. Poster – **Zhen Chen**, Krishnan Padmanabhan. Top-down feedback gates flexible coding strategies in olfactory cortex. *Society for Neuroscience (SfN)* 2021, October 2021, virtual.
11. Attendance: Thirty-second Conference on Neural Information Processing Systems (NeurIPS), December 2018, Montreal, Canada.
12. Attendance: Conference on Cognitive Computational Neuroscience (CCN) 2018, September 2018, Philadelphia, Pennsylvania, USA.
13. Invited Talk: The 5th National Conference on Stochastic Dynamics of China, October 2017, Weihai, China
14. Talk: The 16th National Conference on Nonlinear Vibration & The 13th National Conference on Nonlinear Dynamics and Stability, May 2017, Hangzhou, China.
15. Invited Talk: Conference on New Progress of Stochastic Dynamics and Control, September 2016, Xi'an, China,
16. Invited Talk: The 5th National Conference on Stochastic Dynamics of China & The 10th National Conference on Theory and Applications of Random Vibration, November 2016, Fuzhou, China.
17. Talk: The Chinese Conference of Theoretical and Applied Mechanics, August 2015, Shanghai, China,
18. Attendance: The 4th National Conference on Stochastic Dynamics of China, April 2015, Xi'an, China.

19. Attendance: The 9th National Conference on Theory and Applications of Random Vibration & The 3rd National Conference on Stochastic Dynamics, October 2014, Lanzhou, China.

TEACHING

Kaufman Teaching Certificate Program, MIT	Fall 2026
Teaching Assistant – Neural Circuits for Cognition (by Ila Fiete), MIT	Fall 2024
Guest lecturer – Introduction to Computational Neuroscience (by Ralf Haefner), University of Rochester	Fall 2022
Teaching Assistant – Animal Minds (by Dora Biro), gave 3 lectures on topics about sensory systems, learning and memory, University of Rochester	Fall 2021
Lecturer – Advanced Algebra, Nanjing University of Aeronautics and Astronautics	Fall 2015

ADVISING & MENTORING

Alicia Lu, Graduate student, Training large-scale spiking neural network model with empirical NHP data, MIT	2025 – Present
Helen Qian, Rotation student, Temporal relationship between alpha/beta and spiking activity, MIT	2026
Kai Ding, Rotation student, Context-dependent information representation in working memory, MIT	2024 – 2025
Forrest Wargo, Undergraduate student, Optimizing sampling-based probabilistic inference network model, University of Rochester	2021 – 2022

SERVICES

Publicity and Outreach Officer for Westgate Executive Committee, MIT	2024 – Present
Graduate Assistant for Graduate & Family Housing, University of Rochester	2021 – 2023
BCS Community Support Committee, University of Rochester	2021 – 2022